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**PARIS
SCIENCES
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Statistical Learning II

Lecture 3 - Supervised learning (continued)

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Empirical risk

Let $\mathcal{D} = \{(x_i, y_i) \in \mathcal{X} \times \mathcal{Y} : i = 1, \dots, n\}$ denote the **training data**.

Given a loss function $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}_+$, and a predictor $f : \mathcal{X} \rightarrow \mathcal{Y}$ define the **empirical risk**:

$$\hat{\mathcal{R}}_n(f) = \frac{1}{n} \sum_{i=1}^n \ell(y_i, f(x_i))$$

Also known as the **training loss**. This quantifies how well we fit the data. But is this a good notion of learning?

$$f(x) = \begin{cases} y_i & \text{if } x \in \mathcal{D} \\ 0 & \text{otherwise} \end{cases} \quad \Rightarrow \quad \hat{\mathcal{R}}_n = 0$$

Probabilistic framework

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- The “i.i.d.” assumption might not always hold. (Sampling bias, distribution shift, etc.)
- Under this assumption, $\hat{\mathcal{R}}_n$ is a random function.

Population risk

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Define the notion of population risk of a predictor $f: \mathcal{X} \rightarrow \mathcal{Y}$:

$$\mathcal{R}(f) = \mathbb{E} [\ell(y, f(x))]$$

Also known as the **generalisation** or **test error**.

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$\mathcal{R}$ is a deterministic function of the predictor f

Validation set

In practice, the statistician almost never has access to the data distribution.

A common procedure to estimate $\mathcal{R}$ consists of splitting the training data in **training** and **validation** set $\mathcal{D} = \mathcal{D}_T \cup \mathcal{D}_V$.

Train on $\mathcal{D}_T$, test on $\mathcal{D}_V$.

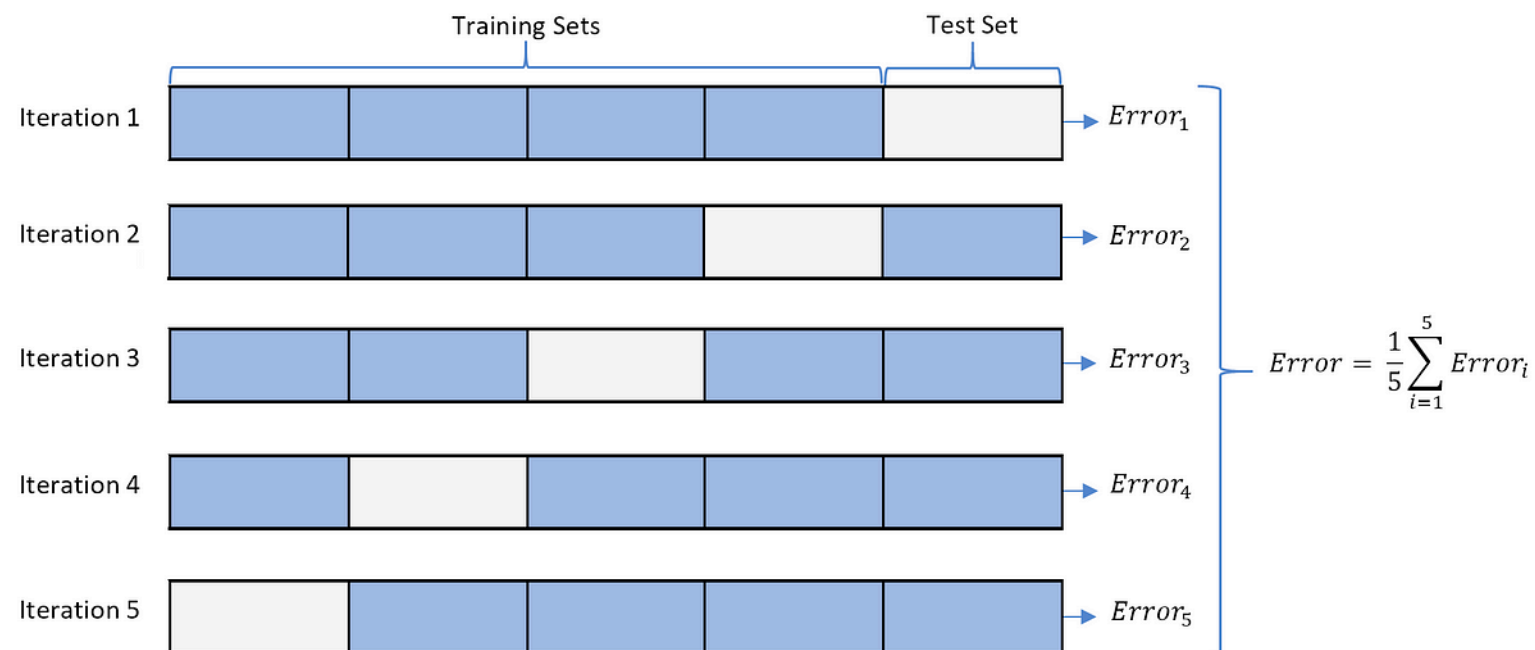
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To reduce error, often one repeats this procedure k times, averaging over the result. This is known as **k fold cross-validation**.



Conditional risk

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The internal expectation is over the conditional distribution $Y|X = x$

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$$\begin{aligned}\mathcal{R}(f) &= \mathbb{E}_{(X,Y) \sim p}[\ell(Y, f(X))] \\ &= \mathbb{E}_{\textcolor{red}{X} \sim p_x} \left[r(z \mid \textcolor{red}{X}) \right] \quad z = f(x) \in \mathcal{Y}\end{aligned}$$

Where we have defined:

$$r(z \mid x) = \mathbb{E}[\ell(Y, z) \mid X = x]$$

“Conditional risk”

Bayes risk

The Bayes predictor is the best achievable predictor:

$$f_{\star}(x) \in \operatorname{argmin}_{z \in \mathcal{Y}} r(z | x)$$

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- The Bayes predictor $f_{\star}$ might not be unique.
- Typically we have $\mathcal{R}_{\star} \neq 0$. 🤔 Examples in the TD

Learning algorithm

Let $\mathcal{D}_p = \{(x_i, y_i) \in \mathcal{X} \times \mathcal{Y} : i = 1, \dots, n\}$ denote training data sampled i.i.d. from p .

A **learning algorithm** is a map that takes the training data and returns a predictor

$$\mathcal{A} : \mathcal{D}_p \mapsto f$$

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You have seen many examples in “*Statistical Learning I*”:

- K-nearest neighbours
- Decision trees
- Random Forests
- Least-squares regression

No free lunch

Consider a binary classification task with $\mathcal{Y} = \{0,1\}$ and 0/1 loss $\ell(y, z) = \delta_{yz}$. Let $\mathcal{P}$ denote the set of probability distributions over $\mathcal{X} \times \{0,1\}$.

Theorem

For any $n \in \mathbb{N}$ and algorithm $\mathcal{A}$ over $(\mathcal{X} \times \{0,1\})^{\otimes n}$, there exists $p \in \mathcal{P}$ such that

$$\sup_{p \in \mathcal{P}} \left\{ \mathbb{E} \left[\mathcal{R}(\mathcal{A}(\mathcal{D}_p)) \right] - \mathcal{R}_\star \right\} \geq \frac{1}{2}$$

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In words: For any algorithm you choose, one can always construct a data distribution such that your error is at best equal than random guessing.

Empirical risk minimisation

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Empirical risk minimisation (ERM) is a class of learning algorithms that consist of minimising the empirical risk:

$$\min_f \frac{1}{n} \sum_{i=1}^n \ell(y_i, f(x_i)) \quad (= \hat{\mathcal{R}}_n(f))$$

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By the law of large numbers, for a given f

$$\hat{\mathcal{R}}_n(f) \xrightarrow{P} \mathcal{R}(f) \quad \text{as} \quad n \rightarrow \infty$$

However, at fixed n , $\hat{\mathcal{R}}_n$ can be very different from $\mathcal{R}$

Empirical risk minimisation

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ERM maps supervised learning to an **optimisation problem**.

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But optimising on the space of functions is computationally intractable....

Empirical risk minimisation

$$\min_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \ell(y_i, f(x_i)) \quad (= \hat{\mathcal{R}}_n(f))$$

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But optimising on the space of functions is computationally intractable....

Therefore, we restrict to classes of mathematically and computationally amenable functions:

$$f \in \mathcal{H}$$

Also known as the **hypothesis class**.

Hypothesis class

Most of the time, we consider **parametric classes**.

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The choice of hypothesis (or architecture) induces an **inductive bias** in the learning.

e.g. linear functions can only learn linear relationships

Risk decomposition

For any $\theta \in \Theta$, we can decompose the excess risk:

$$\mathcal{R}(\theta) - \mathcal{R}_\star = \left(\mathcal{R}(\theta) - \inf_{\theta' \in \Theta} \mathcal{R}(\theta') \right) + \left(\inf_{\theta' \in \Theta} \mathcal{R}(\theta') - \mathcal{R}_\star \right)$$

Estimation error

Approximation error

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 - Deterministic
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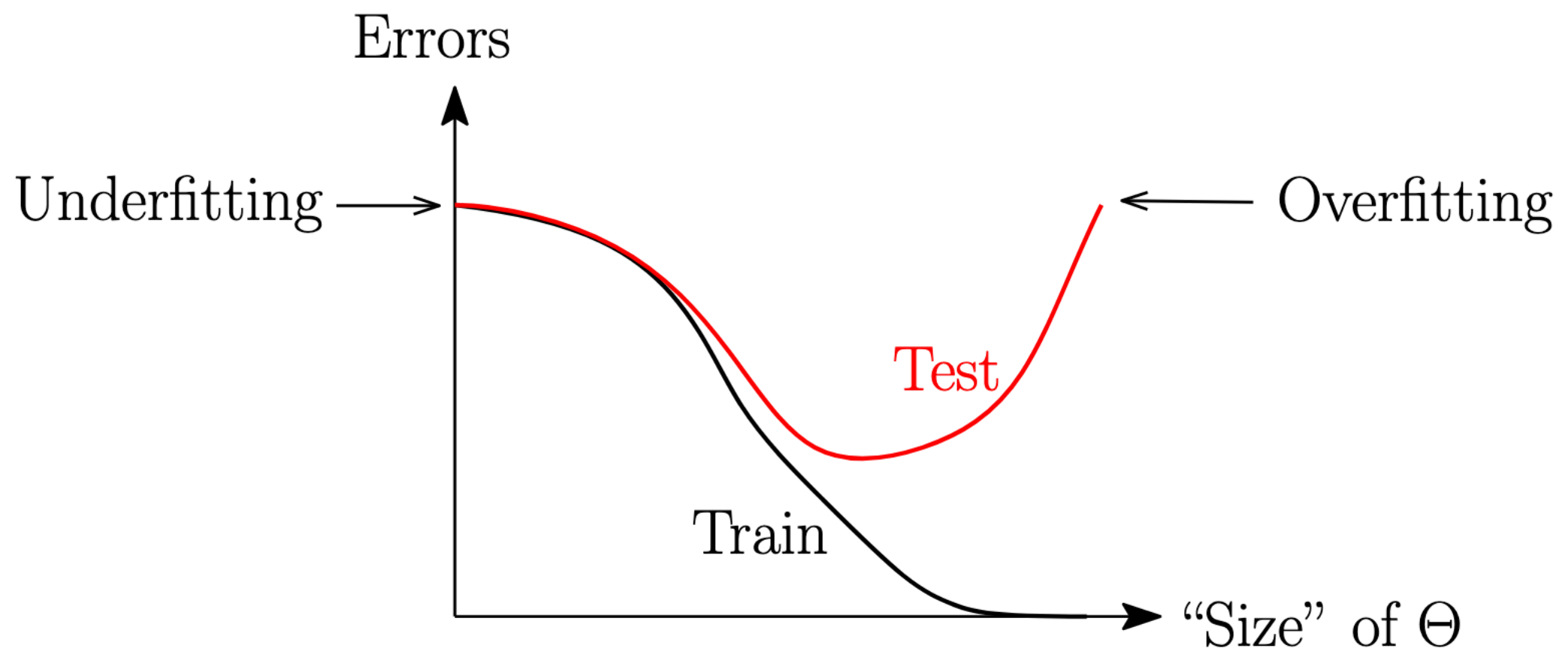


Figure from “Learning Theory from First Principles”, F. Bach 2024

Summary of ERM

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- How large n needs to be (with respect to p, d) so that $\hat{\theta} \in \operatorname{argmin} F(\theta)$ has low training and/or test error?
- What properties of the data distribution p makes the problem easier / harder?